1a.)

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$$

$\frac{1}{2}R1 + R2 \rightarrow R2$

$$\begin{pmatrix} 2(1/2) & 1(1/2) \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 0 & 5/2 \end{pmatrix} = U$$

L-matrix:

$$A_1 = \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2/2 & 0 \\ -1/2 & 2/2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ -1/2 & 1 \end{pmatrix} \rightarrow L = \begin{pmatrix} 1 & 0 \\ -1/2 & 1 \end{pmatrix}$$

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$\xrightarrow{\frac{1}{2}R1 + R2 \to R2}$

$$\begin{pmatrix} 2(1/2) & -1(1/2) & 0(1/2) \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1 \\ 0 & -1 & 2 \end{pmatrix} \xrightarrow{2/3R2 + R3 \to R3}$$

$$\begin{pmatrix} 2 & -1 & 0 \\ 0 & 3/2 & -1/2 \\ 0 & 0 & 4/3 \end{pmatrix} = U$$

$$A_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 2/2 & -1 & 0 \\ -1/2 & 2/2 & -1 \\ 0 & -1/2 & 2/2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & -1 \\ 0 & -1/2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ -1/2 & 1 & -1 \\ 0 & -1/2 & 1 \end{pmatrix} = L$$

L-matrix)

$$A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix}$$ In this form

~~A(6,2)~~ ✓

* for this case normalize diagonal and any underneath do a_{ij}/a_{ii} for bottom entries

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 2/3 & 1/2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 2/3 & 1/2 & 0 \end{pmatrix}$$

~~Upper matrix~~:

$$A_3 = \begin{pmatrix} 3 & 0 & 0 \\ -1 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{\frac{1}{3}R1 + R2 \to R2}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 2 & 1 & 2 \end{pmatrix} \xrightarrow{-\frac{2}{3}R1 + R3 \to R3}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow{-\frac{1}{2}R2 + R3 \to R3}$$

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = U$$